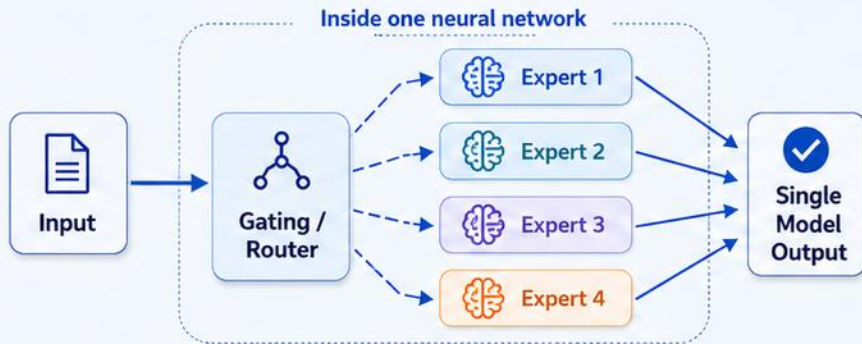


Mixture of Experts vs Mixture of Agents

Two ways to use multiple specialists to improve AI outputs

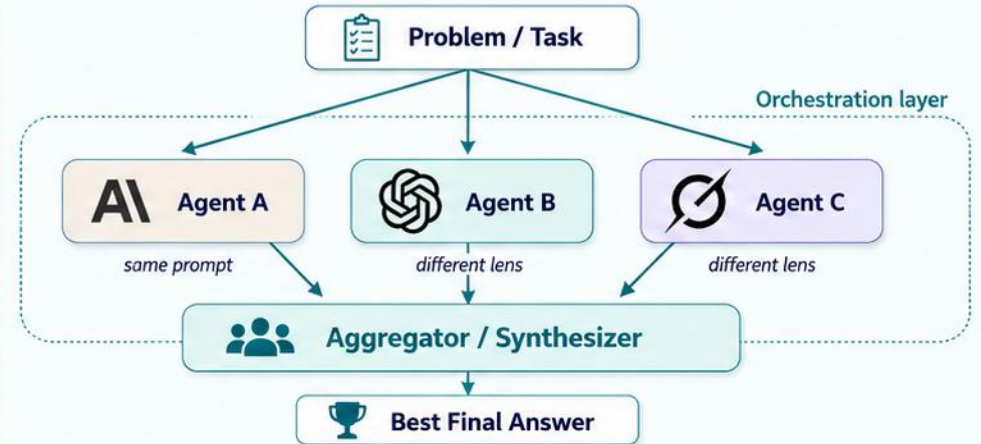
Mixture of Experts (MoE)



-  Neural network architecture
-  Specialized internal expert sub-networks
-  Router activates the most relevant experts
-  Goal: more capacity and efficiency

★ Best understood as a **model design pattern**.

Mixture of Agents (MoA)



-  Multi-agent orchestration pattern
-  Independent LLM agents generate candidate responses
-  Agents can share the same prompt or use different perspectives
-  Aggregator synthesizes the strongest final answer

★ Best understood as a **workflow and coordination pattern**.

Why Mixture of Agents is Promising



Broader coverage



Reduced blind spots



Higher-confidence outputs



Better for complex tasks like coding and security



Enables patterns like ensemble, specialized lenses, critique, and debate



Simple takeaway:

MoE improves the model from the **inside**.

MoA improves the answer from the **outside**.

The Core Distinction

Mixture of Experts improves the model from the **inside**. Mixture of Agents improves the answer from the **outside**.

 Dimension	 Mixture of Experts	 Mixture of Agents
 What it is	Neural network architecture	Orchestration pattern
 Where it happens	Inside the model	Outside the model
 How it works	A router activates internal expert networks	Multiple agents generate outputs that are aggregated
 Who controls it	Model builder	Application, platform, or workflow designer
 Primary goal	Scale model capacity efficiently	Improve answer quality, coverage, and review
 Final output	One model output	One synthesized answer

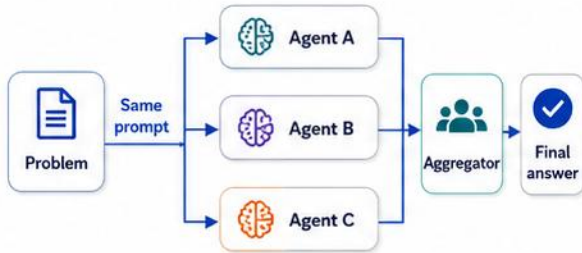


MoE is model architecture. **MoA** is workflow orchestration.

5 Common Sub-Patterns of Mixture of Agents (MoA)

Practical orchestration patterns for combining multiple LLM agents to improve quality, coverage, and robustness

1. Ensemble / Consensus



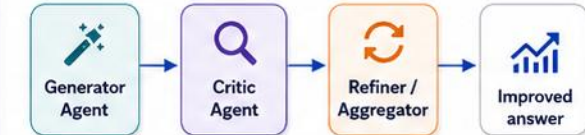
- ✓ Uses model diversity for robustness
- ✓ Good for reliability and confidence
- ✓ Best when the task has a mostly correct answer

2. Role-Based / Perspective Decomposition



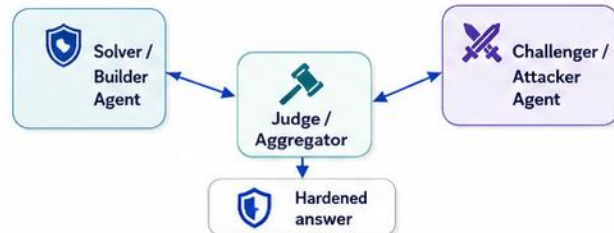
- ✓ Forces viewpoint diversity
- ✓ Improves coverage across tradeoffs
- ✓ Useful for complex, multi-dimensional decisions

3. Generate → Critique → Refine



- ✓ Adds a review loop
- ✓ Strong for coding and design quality
- ✓ Helps catch flaws before final output

4. Red Team vs Blue Team



- ✓ Surfaces vulnerabilities and edge cases
- ✓ Useful for security and policy testing
- ✓ Introduces adversarial robustness

5. Decompose → Solve → Combine



- ✓ Breaks large problems into parts
- ✓ Scales well for complex workflows
- ✓ Useful for multi-step engineering tasks



Simple takeaway

Mixture of Agents is not one pattern. It is a family of orchestration strategies that combine multiple agents, perspectives, or review loops to produce a stronger final answer.

